**Title:** Radar Satellite Imagery Environmental Change Detection Report

**Alert:** LOW

**Date Issued:** 3rd April,2019

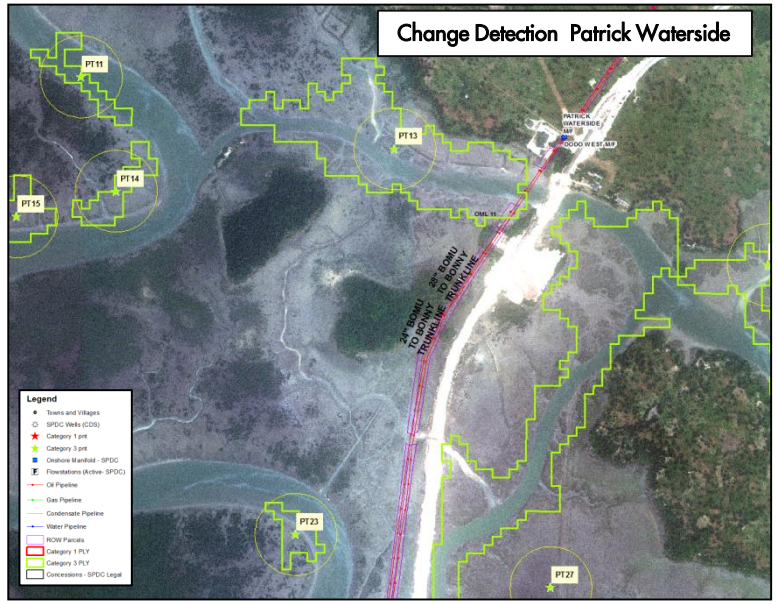
**Geonatics Team: Remote Sensing Analyst**: Oyetunde Simisola, **Reviewed By:** Obowu Chituru, **Head Geoinformation Management**: Bisike-Ojiako Chuks

**Recipients:** Ojiako Uche UPO/G/UPR, Unutaro Tobore Grace UPO/G/UPR

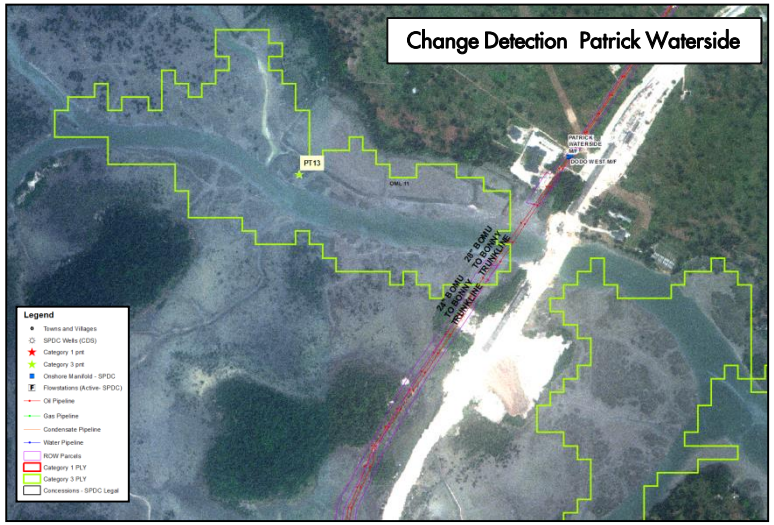
List of Change Detection Points and Attributes

| S/N | Northings (Minna/Nigeria Mid Belt) | Eastings (Minna/Nigeria Mid Belt) | Longitude (WGS84) | Latitude (WGS84) | Alert Category | Alert Level | Change Detection Date | Remarks |
|---|---|---|---|---|---|---|---|---|
| PT 11 | 67651.96 | 531642.37 | 7.24728 | 4.61152 | Category 3 | Low | 20190305-20190329 | Changes of water coverage (e.g. tidal, river, flood) as well as wetland related changes |
| PT 13 | 67388.42 | 532791.86 | 7.257644 | 4.609155 | Category 3 | Low | 20190305-20190329 | Changes of water coverage (e.g. tidal, river, flood) as well as wetland related changes |
| PT 14 | 67232.77 | 531770.69 | 7.248444 | 4.607731 | Category 3 | Low | 20190305-20190329 | Changes of water coverage (e.g. tidal, river, flood) as well as wetland related changes |
| PT 15 | 67139.51 | 531406.53 | 7.245163 | 4.606882 | Category 3 | Low | 20190305-20190329 | Changes of water coverage (e.g. tidal, river, flood) as well as wetland related changes |
| PT 17 | 66963.37 | 534161.19 | 7.269991 | 4.605333 | Category 3 | Low | 20190305-20190329 | Changes of water coverage (e.g. tidal, river, flood) as well as wetland related changes |
| PT 21 | 66399.65 | 534480.79 | 7.272881 | 4.60024 | Category 3 | Low | 20190305-20190329 | Changes of water coverage (e.g. tidal, river, flood) as well as wetland related changes |
| PT 23 | 65974.88 | 532429.09 | 7.254397 | 4.596366 | Category 3 | Low | 20190305-20190329 | Changes of water coverage (e.g. tidal, river, flood) as well as wetland related changes |
| PT 27 | 65780.41 | 533364.42 | 7.262829 | 4.594622 | Category 3 | Low | 20190305-20190329 | Changes of water coverage (e.g. tidal, river, flood) as well as wetland related changes |

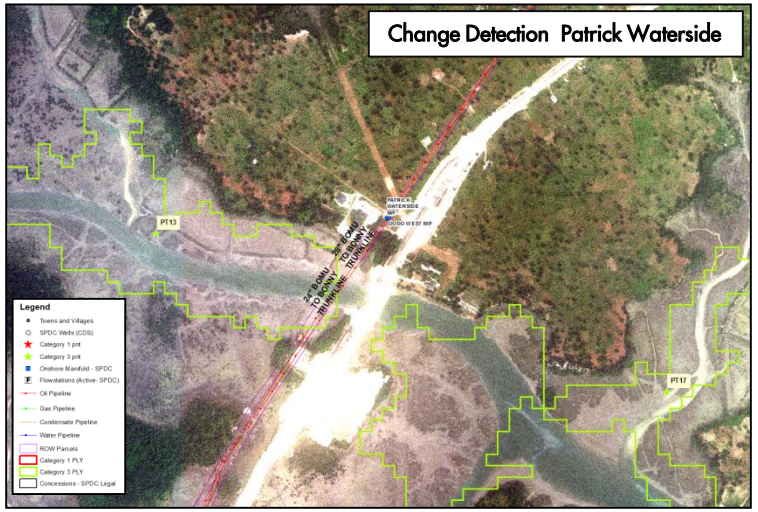
***Note:*** *Points are within 150m sensitivity zone of the centroid of the identified change areas.*




*Clip 1: Full view of change detection*




*Clip 2: View of pt13 change detection*




*Clip 3: View of pt13 &17 change detection*

<u>Summary</u>

**Concession** : OML 11

**Local Government**: Bonny

**Settlements**: NTR

**Wells**: Bodo West-004, Bodo West-003

**Flow Stations**: Nil

**Manifolds**: Patrick Waterside M/F, Dodo North M/F, Bodo West M/F

**Pipelines**: 24" Bomu to Bonny Trunk line, 28" Bomu to Bonny Trunk line

<u>Alert Category Description</u>

- **Category 1 Targets** - metallic man-made features such as vehicle movement, ship traffic or structures.
- **Category 2 Canopy/Surface** - tree canopy changes/ disturbance) and surface disturbance.
- **Category 3 Low Backscatter** - e.g. tidal, river, flood) as well as wetland related changes.

<u>Remote sensing Analysis Result</u>

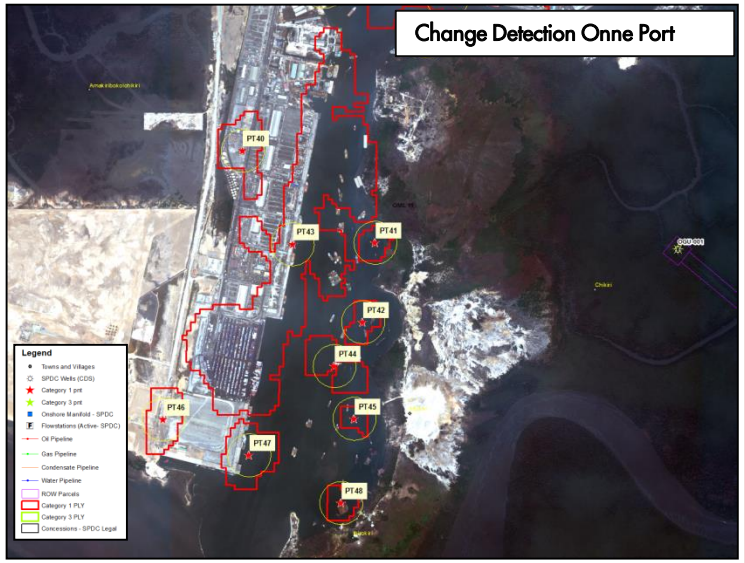
- Possible changes are within 150m radius of detected alert points as analyzed by radar satellite imagery acquired on 05/03/2019 and 29/03/2019.
- Category 3 changes identified within this area. Changes possibly related to tidal, river, flooding as well as wetland related changes.
- All points are category 3 change which is most likely changes in water coverage (e.g. tidal, river, flood) as well as wetland related changes.
- All alert level are low.
- No action required.

| **Title:** Radar Satellite Imagery Environmental Change Detection Report | | **Alert:** LOW |
|---|---|---|
| **Date Issued:** 3rd April,2019 | **Geonatics Team: Remote Sensing Analyst**: Oyetunde Simisola, **Reviewed By:** Obowu Chituru, **Head Geoinformation Management**: Bisike-Ojiako Chuks | **Recipients:** Ojiako Uche UPO/G/UPR, Unutaro Tobore Grace UPO/G/UPR |

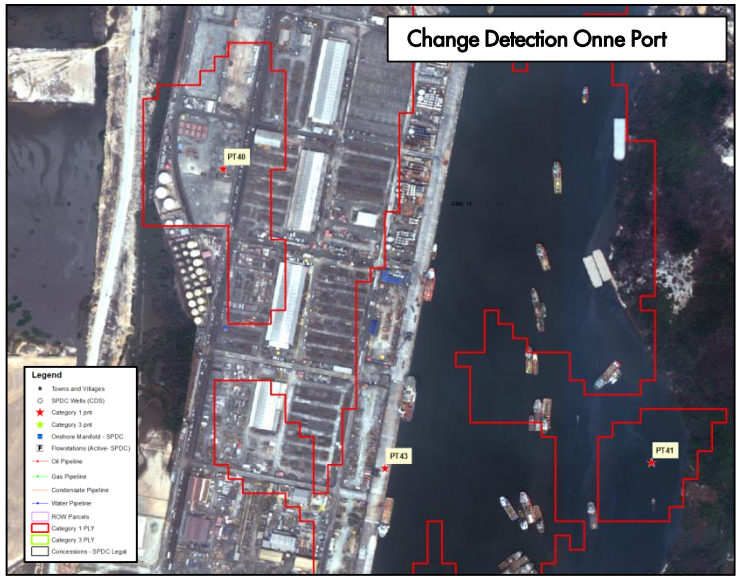
## List of Change Detection Points and Attributes

| S/N | Northings (Minna/Nigeria Mid Belt) | Eastings (Minna/Nigeria Mid Belt) | Longitude (WGS84) | Latitude (WGS84) | Alert Category | Alert Level | Change Detection Date | Remarks |
|---|---|---|---|---|---|---|---|---|
| PT 40 | 75850.79 | 533755.61 | 7.152743 | 4.685487 | Category 1 | Low | 20190305-20190329 | Vehicle movement, ship ,changes to or the addition of new structures. |
| PT 41 | 75208.79 | 522095.13 | 7.16111 | 4.679698 | Category 1 | Low | 20190305-20190329 | Vehicle movement, ship ,changes to or the addition of new structures. |
| PT 42 | 74653.68 | 522008.94 | 7.160343 | 4.674676 | Category 1 | Low | 20190305-20190329 | Vehicle movement, ship ,changes to or the addition of new structures. |
| PT 43 | 75197.75 | 521517.85 | 7.155908 | 4.679588 | Category 1 | Low | 20190305-20190329 | Vehicle movement, ship ,changes to or the addition of new structures. |
| PT 44 | 74346.83 | 521809.70 | 7.158553 | 4.671898 | Category 1 | Low | 20190305-20190329 | Vehicle movement, ship ,changes to or the addition of new structures. |
| PT 45 | 73978.88 | 521949.04 | 7.159815 | 4.668573 | Category 1 | Low | 20190305-20190329 | Vehicle movement, ship ,changes to or the addition of new structures. |
| PT 46 | 73972.59 | 520613.31 | 7.147777 | 4.668493 | Category 1 | Low | 20190305-20190329 | Vehicle movement, ship ,changes to or the addition of new structures. |
| PT 47 | 73723.70 | 521213.69 | 7.153192 | 4.666253 | Category 1 | Low | 20190305-20190329 | Vehicle movement, ship ,changes to or the addition of new structures. |
| PT 48 | 73389.66 | 521856.09 | 7.158988 | 4.663243 | Category 1 | Low | 20190305-20190329 | Vehicle movement, ship ,changes to or the addition of new structures. |

*Note: Points are within 150m sensitivity zone of the centroid of the identified change areas.*




*Clip 1: Full view of change detection*




*Clip 2: View of Pt40,Pt41,Pt43 change detection*

**Summary**

Concession :OML11

Local Government: Ogu/Bolo

Settlements: Ibirokiri, Chikiri, Uwogolo, Amakiribokoiohikiri, Amakiriboko

Wells: Ogu-001

Flow Stations: Nil

Manifolds: Nil

Pipelines: Nil

**Alert Category Description**

- **Category 1 Targets** - metallic man-made features such as vehicle movement, ship traffic or structures.
- **Category 2 Canopy/Surface** - tree canopy changes/ disturbance) and surface disturbance.
- **Category 3 Low Backscatter** - e.g. tidal, river, flood) as well as wetland related changes.

**Remote sensing Analysis Result**

- Possible changes are within 150m radius of detected alert points as analyzed by radar satellite imagery acquired on 05/03/2019 and 29/03/2019.
- Category 1 changes identified within this area. Changes possibly related to vehicle movement, ship, changes to or the addition of new structures .This area is not within our primary area of interest. But alert is for information only.
- All alert level are low.
- No action required.

*Clip 2: View of Pt42,Pt44,Pt45 change detection*